

The ordinary Kreweras complement: an exact finite cycle atlas

HCS-C209 source-derived research package

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Abstract

We freeze a finite dynamical system on the noncrossing partitions of a labelled polygon. The clock is one ordinary Kreweras complement, defined by a permutation formula with an explicit orientation. The type-A order- $2n$ cyclic-sieving theorem supplies every iterate fixed count. We then derive, without fitting, all least-period populations, cycle counts, the finite Artin–Mazur zeta, the reciprocal Koopman determinant, root-of-unity spectral multiplicities, rank duality, and polygon-reflection reversors. The actual order is 1, 2, or $2n$ at $n = 1$, $n = 2$, or $n \geq 3$; this small-dimensional boundary is part of the theorem. Independent set-partition enumeration through $n = 8$ and exact SymPy cyclotomic checks through $n = 12$ audit the ledger. The result is deliberately source-local: Catalan data carry no intrinsic rational-prime clock, so no arithmetic determinant or Hilbert–Pólya claim is made.

1 Object and orientation

Place vertices $[n] = \{0, 1, \dots, n-1\}$ counterclockwise on a convex polygon. A set partition is noncrossing when no two blocks contain a, c and b, d in the cyclic order $a < b < c < d$. For $\pi \in \text{NC}(n)$, let p_π cycle each block in increasing order, and let $c = (0\ 1\ \dots\ n-1)$. We use

$$K(\pi) = \text{cycles}(p_\pi^{-1}c). \quad (1)$$

The notation cycles means the set partition formed by the cycles of a permutation. This convention gives the clockwise rotation $\rho_{-1}(i) = i-1 \pmod{n}$ in the square identity below. Replacing c by c^{-1} conjugates the action by a reflection and changes only this sign.

There are two useful orders. Let

$$L_n = \begin{cases} 1, & n = 1, \\ 2, & n = 2, \\ 2n, & n \geq 3, \end{cases} \quad G_n = \begin{cases} 1, & n = 1, \\ 2n, & n \geq 2. \end{cases} \quad (2)$$

L_n is the actual order of the permutation K . G_n is the abstract cyclic-group order in the order- $2n$ CSP. At $n = 2$ the action has a kernel of order two, so these symbols must not be conflated.

2 Structural identities

Theorem 2.1 (ordinary complement identities). *For every n , K is a bijection of $\text{NC}(n)$ and*

$$K^2 = \rho_{-1}, \quad |K(\pi)| = n + 1 - |\pi|. \quad (3)$$

It reverses refinement. Every reflection $R_j(i) = j - i \pmod{n}$ obeys $R_j^2 = 1$ and $R_j K R_j = K^{-1}$. The actual order is L_n in (2).

Proof sketch and boundary check. The permutation factorization $p_\pi p_{K(\pi)} = c$ is the usual geodesic factorization for a noncrossing partition. Taking inverses and multiplying by c gives the complement involution on the geodesic interval, hence the block count and refinement reversal. Applying the factorization twice relabels all vertices by $i \mapsto i - 1$, giving the first identity in (3). Conjugating the displayed permutations by R_j gives the dihedral relation. For $n \geq 3$, polygon rotation acts faithfully on $\text{NC}(n)$, so K^2 has order n . Every odd power reverses rank and therefore sends the discrete partition to the indiscrete partition; in particular K^n is not the identity. Thus the order is $2n$. Directly, $\text{NC}(1)$ is a singleton and K exchanges the two elements of $\text{NC}(2)$. $\square$

The rank $r(\pi) = n - |\pi|$ has Narayana distribution

$$\#\{\pi : |\pi| = b\} = \frac{1}{n} \binom{n}{b} \binom{n}{b-1}, \quad |\text{NC}(n)| = \text{Cat}_n = \frac{1}{n+1} \binom{2n}{n}. \quad (4)$$

The rank involution in Theorem 2.1 is checked row by row in the machine-readable evidence.

3 The fixed-point theorem

Write $[m]_q = 1 + q + \cdots + q^{m-1}$ and

$$\text{Cat}_n(q) = \frac{[2n]_q!}{[n]_q![n+1]_q!} \in \mathbb{Z}[q]. \quad (5)$$

The rotation CSP for $\text{NC}(n)$ is the type-A theorem of Reiner–Stanton–White. The order- $2n$ complement extension is the $m = 1$ type-A Kreweras CSP, recorded by Bessis–Reiner and credited there to direct calculations of White. We use these results with attribution, not as a claim of priority.

Theorem 3.1 (all-iterate fixed ledger). *Let $F_n(d) = |\text{Fix}(K^d)|$. For $n = 1$, $F_1(d) = 1$. For $n \geq 2$, reduce d modulo L_n . If $d = 2r$ is even, then*

$$F_n(2r) = \begin{cases} \text{Cat}_n, & r \equiv 0 \pmod{n}, \\ \binom{2g}{g}, & 0 < r < n, \quad g = \gcd(n, r). \end{cases} \quad (6)$$

For odd d , $F_n(d) = \binom{n}{(n-1)/2}$ exactly when n is odd and $d \equiv n \pmod{2n}$; all other odd rows are zero. Equivalently, for $n \geq 2$,

$$F_n(d) = \text{Cat}_n(e^{2\pi i d / G_n}). \quad (7)$$

At $n = 2$, equation (7) uses fourth roots for the abstract C_4 action even though the image permutation has order two.

Root reduction. The CSP gives (7). For even exponents it is the usual rotation specialization: a rotation with $g = \gcd(n, r)$ fixes $\binom{2g}{g}$ partitions, with the identity giving Cat_n . At an odd root the cyclotomic factors in (5) cancel only for the central half-turn; rank reversal also forces $|\pi| = (n+1)/2$, so this row is absent for even n and has the central binomial value for odd n . These elementary cancellations are implemented with integer polynomial remainders in the SymPy audit. $\square$

For orientation, the first rows are

n	L_n	Cat_n	$(F_n(0), F_n(1), \dots, F_n(L_n - 1))$
1	1	1	(1)
2	2	2	(2,0)
3	6	5	(5,0,2,3,2,0)
4	8	14	(14,0,2,0,6,0,2,0)
5	10	42	(42,0,2,0,2,10,2,0,2,0)
6	12	132	(132,0,2,0,6,0,20,0,6,0,2,0)

4 Periods, zeta, and Koopman spectrum

The rest of the atlas is a finite consequence and needs no additional source assumption. Let μ denote the integer Möbius function.

Theorem 4.1 (cycle and determinant package). *For every $\ell \mid L_n$, define*

$$P_{n,\ell} = \sum_{d \mid \ell} \mu(\ell/d) F_n(d), \quad C_{n,\ell} = P_{n,\ell}/\ell. \quad (8)$$

Then $P_{n,\ell}$ is the number of points of least period ℓ and $C_{n,\ell}$ is the number of cycles. They are nonnegative integers and sum to Cat_n . The finite zeta and Koopman determinant are

$$\zeta_{K,n}(z) = \prod_{\ell \mid L_n} (1 - z^\ell)^{-C_{n,\ell}}, \quad \det(I - zU_n) = \prod_{\ell \mid L_n} (1 - z^\ell)^{C_{n,\ell}} = \zeta_{K,n}(z)^{-1}, \quad (9)$$

where $U_n f = f \circ K$ on $\ell^2(\text{NC}(n))$.

Proof. The fixed-point trace identity is $F_n(d) = \sum_{\ell \mid d} P_{n,\ell}$. Möbius inversion gives (8); dividing by ℓ counts cycles. Each cycle contributes the factor $1 - z^\ell$ to a permutation determinant and its reciprocal to the Artin–Mazur product. This proves (9). $\square$

Corollary 4.2 (spectrum and reversor). *The spectrum of U_n consists of L_n th roots, with*

$$\text{mult}\left(e^{2\pi i k/L_n}\right) = \sum_{\substack{\ell \mid L_n \\ L_n \mid k\ell}} C_{n,\ell}, \quad \text{Tr}(U_n^d) = F_n(d). \quad (10)$$

The permutation matrix is unitary. For every j , the map $J_j f = \overline{f \circ R_j}$ is antiunitary and reverses U_n : $J_j U_n J_j = U_n^{-1}$.

For example, the nonzero cycle lengths for $n = 3, 4, 5, 6$ are respectively $\{2, 3\}$, $\{2, 4, 8\}$, $\{2, 5, 10\}$, and $\{2, 4, 6, 12\}$. The machine ledger stores every exponent rather than only these summaries.

5 Independent evidence

The producer emits exact rows for $1 \leq n \leq 24$ (597 fixed rows, 131 period rows, 597 spectral rows, and 300 rank rows). It also emits q-Catalan coefficients through $n = 12$. A separate checker reimplements restricted-growth set partitions, crossing detection, the permutation in (1), all reflections, and the cycle decomposition. It enumerates all 2,055 partitions for $1 \leq n \leq 8$ and reports 7,808 assertions. It does not import producer code. A separate SymPy program reconstructs the

q-factorial quotient and cyclotomic remainders (1,110 exact checks), then independently recomputes the period and spectral rows through $n = 24$.

The byte-replay script regenerates identical JSON. The mutation harness tests 31 repaired-hash payload mutations and one stale-hash mutation; all are rejected. The release manifest excludes LaTeX sidecars and itself, and stores the hashes of exactly 27 payload files, including the three round PDFs.

6 Scope, relation to C187, and conclusion

C209 is a complete finite combinatorial atlas, not an arithmetic promotion. The integer n , Catalan/Narayana counts, and polygon roots have no intrinsic rational-prime carrier. We therefore record the conservative evaluator tuple

(A0_FAIL, A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION),

with overall verdict ‘ROUTE_A_REJECTED’ and no Route-B invocation. The only operator is the finite source Koopman permutation; no target divisor, functional equation, Euler factor, root number, automorphy statement, or Hilbert–Pólya operator is asserted.

C187 instead owns promotion of rectangular standard Young tableaux, its q-hook polynomial, and evacuation. C209 owns set partitions, geometric complement, the K^2 polygon rotation, and the order- $2n$ complement clock. Shared Möbius/CSP bookkeeping does not make one a parameter continuation of the other.

References

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- [3] D. Bessis and V. Reiner, “Cyclic sieving of noncrossing partitions for complex reflection groups,” *Annals of Combinatorics* 15(2) (2011), 197–222.